



THE GREAT SORTING CHALLENGE: ORGANIZING OUR WORLD

THE MESSY LIBRARY

Imagine walking into a library where books are scattered everywhere on tables, floors, and shelves! You need to find "Wings of Fire" by APJ Abdul Kalam for your project.

How do we quickly find the right book?

Searching randomly takes forever and is frustrating!

We need order — sorting helps us find things fast!



WHAT IS SORTING?

Sorting means arranging things in a specific order — like lining up from shortest to tallest! Think of your morning assembly where everyone stands in a neat line based on their height. That's sorting in action!

Sorting helps us organize and find things quickly and easily.



BUBBLE SORT — THE NEIGHBOR SWAP

Bubble Sort means swapping neighbors if they are in the wrong order. Imagine sorting cricket bats by length — compare two at a time and swap if needed!

Compare two cricket bats next to each other and swap if the first is longer.

Let's Do It: Imagine swapping cricket bats until all are in order!



BUBBLE SORT STEP-BY-STEP

Follow these simple steps to sort cricket bats from shortest to longest using Bubble Sort!



Step 1: Look at two cricket bats next to each other.

Step 2: If the first is longer than the second, swap them!

Step 3: Move to the next pair and repeat the same check.

Step 4: Keep going until no swaps are needed — done!

SELECTION SORT — THE SMALLEST FIRST

Selection Sort means finding the smallest item and putting it at the start. Imagine picking mangoes from a basket — always choose the smallest one first!

Pick the smallest mango from the basket and move it to a new box.

Let's Do It: Pick the smallest mango each time until all are sorted!



SELECTION SORT STEP-BY-STEP

Follow these simple steps to sort mangoes from a basket into a new box, smallest first!

Look through the whole basket to find the smallest mango.

Repeat with the rest of the mangoes.

Put that mango in the new box at the front.

Continue until all mangoes are moved in order.



MERGE SORT — THE TEAM SPLIT

Merge Sort uses a clever trick: Break the big group into smaller groups, sort each one, then join them back together!

Two friends sort their own card collections, then merge into one big deck.

Divide and Conquer makes big problems easier to solve!



HOW MERGE SORT WORKS

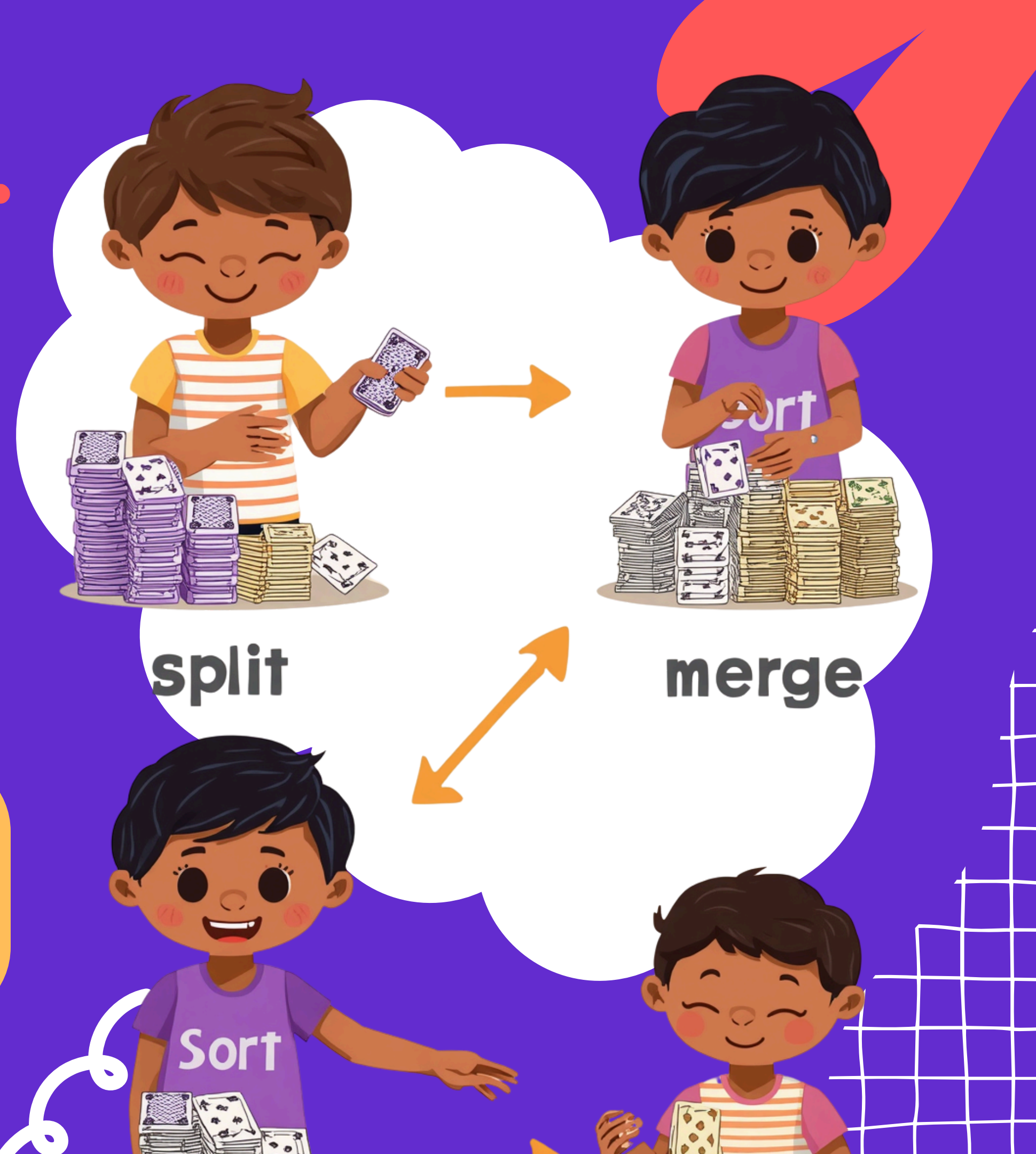
Split, Sort, and Merge!

Merge Sort uses a simple three-step process to organize items. Let's see how it works with a deck of cards!

Split: Divide the cards into two equal halves.

Sort: Arrange each half in order separately.

Merge: Combine both sorted halves into one sorted deck.



SORTING COMPARISON

Which Method Works Best?

Let's compare all three sorting algorithms we learned today and see when to use each one!

Bubble Sort: Easy neighbor swaps, great for small sets like cricket bats.

Selection Sort: Finds the smallest each time, careful picking like mangoes.

Merge Sort: Best for big sets, splits and joins like card collections.



WHICH SORTING IS BEST?

Think about it! Which sorting method would you choose for different situations? Discuss with your classmates!

A small box of crayons? (Hint: Easy swaps!)

A big school database? (Hint: Split and merge!)



RECAP — WHAT WE LEARNED TODAY

Sorting means arranging things in order — like shortest to tallest. It helps us find things quickly in daily life!

Bubble Sort swaps neighbors until sorted.

Selection Sort picks the smallest first.

Merge Sort splits, sorts, then joins back.



QUIZ TIME!

Let's test what you've learned about sorting algorithms! Answer these questions to show your sorting skills.

Q1: What does Bubble Sort do with two neighbors?

Q3: What does Merge Sort do before joining groups?

Q2: In Selection Sort, what do you look for first?

Write your answers or discuss with your classmates!



QUIZ TIME!

QUESTIONS 4-5

Test your sorting knowledge! Think carefully and share your answers with the class or write them down.

1. Which sorting method is best for a small collection of crayons?

2. Why do we sort things in real life?





THANK YOU & KEEP SORTING!

The Great Sorting Challenge Complete!

Thank you for joining The Great Sorting Challenge! You have learned three amazing ways to organize things — Bubble Sort, Selection Sort, and Merge Sort. Now you can sort like a pro!

Try sorting your books, toys, or crayons at home using these fun methods!